

Analysis

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Analysis

Packages

```
set.seed(157)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(glue)
library(zoo)

##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

Functions Script

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
source('inR/Functions.R')
```

Data Sets

Loading in the data sets that I'll need for analysis.

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
team_box_scores <- readRDS(
  'Data-NBA/Current/team_box_scores_current.rds')
player_box_scores <- readRDS(
  'Data-NBA/Current/full_box_player_current.rds')
team_season_stats <- readRDS(
```

```

'Data-NBA/Current/team_season_stats_current.rds')
player_season_stats <- readRDS(
'Data-NBA/Current/player_season_stats_current.rds')

```

Applying a function I wrote that fixes some aesthetic issues with player names. This will occur with foreign players that have special characters or foreign characters in their name.

```

player_box_scores <- fixplayername(player_box_scores)
player_season_stats <- fixplayername(player_season_stats)

```

Outcomes and Small Metrics

Outcome I'm adding the points from the opposing team so I can create an outcome variable based on the margin of the outcome. This will be multinomial regression with four different classifications. Those are...

```

opponent_stats <- team_box_scores %>%
  select(gamedate, team, pts) %>%
  rename('opp_pts'='pts')

team_box_scores <- team_box_scores %>%
  left_join(opponent_stats, by = c('gamedate', 'opponent'='team'))

team_box_scores <- team_box_scores %>%
  mutate(
    ptsdiff = pts - opp_pts,
    twoleveloc = if_else(offrating > defrating, 1, 0),
    threeleveloc = case_when(
      ptsdiff >= 11 ~ 2,
      ptsdiff %in% 1:10 ~ 1,
      TRUE ~ 0),
    fourleveloc = case_when(
      ptsdiff <= -11 ~ 0,
      ptsdiff %in% -1:-10 ~ 1,
      ptsdiff %in% 1:10 ~ 2,
      ptsdiff >= 11 ~ 3)) %>%
  mutate(
    twoleveloc = factor(twoleveloc,
      levels = c(0, 1),
      labels = c('Loss', 'Win')),
    threeleveloc = factor(threeleveloc,
      levels = c(0, 1, 2),
      labels = c('Loss', 'Basic Win', 'Blowout Win')),
    fourleveloc = factor(fourleveloc,
      levels = c(0, 1, 2, 3),
      labels = c('Blowout Loss', 'Basic Loss',
        'Basic Win', 'Blowout Win'))))

```

Conditional Outcomes I want to look at how missing a player affects the chance of a team to win a game; however, I have to determine which players are important for the outcome of the game. I'm going to look at few metrics and use those to determine a player's significance. For the different metrics, different players may end up being the most important player. Those metrics include...

1. Win Shares: `ws`
2. Value Over Replacement Player (VORP): `vorp`
3. Player Efficiency Rating: `player_effrtg`

I'm also going to create a variable that has a unique id for each best player by a given metric.

```
bp_by_ws <- best_player_metric(player_season_stats, ws, 20, 10) %>%
  rename('ws_name'='name') %>%
  unite('ws_team_id', c('year', 'team', 'ws_player_id'),
        sep = '_', remove = FALSE) %>%
  mutate(ws_team_id = paste0('ws_', ws_team_id)) %>%
  relocate(ws_team_id, .after = last_col())
bp_by_vorp <- best_player_metric(player_season_stats, vorp, 20, 10) %>%
  rename('vorp_name'='name') %>%
  unite('vorp_team_id', c('year', 'team', 'vorp_player_id'),
        sep = '_', remove = FALSE) %>%
  mutate(vorp_team_id = paste0('vorp_', vorp_team_id)) %>%
  relocate(vorp_team_id, .after = last_col())
bp_by_per <- best_player_metric(player_season_stats, player_effrtg, 20, 10) %>%
  rename('per_name'='name') %>%
  unite('per_team_id', c('year', 'team', 'player_effrtg_player_id'),
        sep = '_', remove = FALSE) %>%
  mutate(per_team_id = paste0('per_', per_team_id)) %>%
  relocate(per_team_id, .after = last_col())
```

Here I'm going to create a variable within each data set that will allow to check for player availability when coding a future outcome variable

Joining the data from the best player by a certain metric data sets to later determine if player was on the court and how that affected the outcome.

```
team_box_scores_2 <- team_box_scores %>%
  left_join(bp_by_ws, by = c('team', 'season'='year')) %>%
  left_join(bp_by_vorp, by = c('team', 'season'='year')) %>%
  left_join(bp_by_per, by = c('team', 'season'='year')) %>%
  mutate(
    ws_tm_game_id = paste0(ws_team_id, '_', gamedate),
    vorp_tm_game_id = paste0(vorp_team_id, '_', gamedate),
    per_tm_game_id = paste0(per_team_id, '_', gamedate))

player_box_scores_2 <- player_box_scores %>%
  left_join(bp_by_ws, by = c('team', 'season'='year', 'name'='ws_name')) %>%
  left_join(bp_by_vorp, by = c('team', 'season'='year', 'name'='vorp_name')) %>%
  left_join(bp_by_per, by = c('team', 'season'='year', 'name'='per_name')) %>%
  mutate(
    ws_plyr_game_id = if_else(
      !is.na(ws_team_id), paste0(ws_team_id, '_', gamedate), NA),
    vorp_plyr_game_id = if_else(
      !is.na(vorp_team_id), paste0(vorp_team_id, '_', gamedate), NA),
    per_plyr_game_id = if_else(
      !is.na(per_team_id), paste0(per_team_id, '_', gamedate), NA))
```

After creating the data sets necessary to determine whether a team's best player (by a given metric) was available I can now create an outcome variable to check for it. I'm also going to add a variable to determines whether a game was a regular season game or playoff game.

Note: *I know I don't have to use multiple mutate functions; however, at times I like to use multiple as it helps me organize my code as I'm writing it.*

```
team_box_scores_3 <- team_box_scores_2 %>%
  mutate(
```

```

ws_bpa = if_else(ws_tm_game_id %in%
  player_box_scores_2$ws_plyr_game_id, 1, 0),
vorp_bpa = if_else(vorp_tm_game_id %in%
  player_box_scores_2$vorp_plyr_game_id, 1, 0),
per_bpa = if_else(per_tm_game_id %in%
  player_box_scores_2$per_plyr_game_id, 1, 0)) %>%
mutate(
  ws_bpa_multi = case_when(
    twoleveloc == 'Loss' & ws_bpa == 0 ~ 0,
    twoleveloc == 'Loss' & ws_bpa == 1 ~ 1,
    twoleveloc == 'Win' & ws_bpa == 0 ~ 2,
    twoleveloc == 'Win' & ws_bpa == 1 ~ 3),
  vorp_bpa_multi = case_when(
    twoleveloc == 'Loss' & vorp_bpa == 0 ~ 0,
    twoleveloc == 'Loss' & vorp_bpa == 1 ~ 1,
    twoleveloc == 'Win' & vorp_bpa == 0 ~ 2,
    twoleveloc == 'Win' & vorp_bpa == 1 ~ 3),
  per_bpa_multi = case_when(
    twoleveloc == 'Loss' & per_bpa == 0 ~ 0,
    twoleveloc == 'Loss' & per_bpa == 1 ~ 1,
    twoleveloc == 'Win' & per_bpa == 0 ~ 2,
    twoleveloc == 'Win' & per_bpa == 1 ~ 3)) %>%
mutate(
  ws_bpa_multi = factor(ws_bpa_multi,
    levels = c(0, 1, 2, 3),
    labels = c('Loss Without', 'Loss With',
      'Win Without', 'Win With')),
  vorp_bpa_multi = factor(vorp_bpa_multi,
    levels = c(0, 1, 2, 3),
    labels = c('Loss Without', 'Loss With',
      'Win Without', 'Win With')),
  per_bpa_multi = factor(per_bpa_multi,
    levels = c(0, 1, 2, 3),
    labels = c('Loss Without', 'Loss With',
      'Win Without', 'Win With')) %>%
mutate(
  gametype = case_when(
    #2021 Season
    (gamedate >= as.Date('2021-05-22') &
      gamedate <= as.Date('2021-07-20')) |
    #2022 Season
    (gamedate >= as.Date('2022-04-16') &
      gamedate <= as.Date('2022-06-16')) |
    #2023 Season
    (gamedate >= as.Date('2023-04-15') &
      gamedate <= as.Date('2023-06-12')) |
    #2024 Season
    (gamedate >= as.Date('2024-04-20') &
      gamedate <= as.Date('2024-06-17')) |
    #2025 Season
    (gamedate >= as.Date('2025-04-19') &
      gamedate <= as.Date('2025-06-2022')) |
    ##2026 Season

```

```
(gamedate >= as.Date('2026-04-18') &
  gamedate <= as.Date('2026-06-13')) ~ 'Playoff Game',
TRUE ~ 'Regular Season Game'))
```

Here I'm going to create some variables that will be used as possible predictors when testing regression models. There are great variables to work with in the data set so far; however, I want to create some rolling averages so I can measure how teams are doing across seasons and within a certain window.

```
regsnbox <- team_box_scores_3 %>%
  filter(gametype == 'Regular Season Game') %>%
  group_by(team) %>%
  arrange(desc(gamedate), .by_group=TRUE) %>%
  mutate(
    roll_fg_pct = rollmean(fg_pct, k=10, fill=NA, align='left'),
    roll_threefg_pct = rollmean(threefg_pct, k=10, fill=NA, align='left'),
    roll_pts = rollmean(pts, k=10, fill=NA, align='left'),
    roll_three_par = rollmean(three_par, k=10, fill=NA, align='left'),
    roll_dreb_pct = rollmean(dreb_pct, k=10, fill=NA, align='left'),
    roll_ftar = rollmean(ftar, k=10, fill=NA, align='left'),
    roll_ast_pct = rollmean(ast_pct, k=10, fill=NA, align='left'),
    roll_offrating = rollmean(offrating, k=10, fill=NA, align='left'),
    roll_defrating = rollmean(defrating, k=10, fill=NA, align='left'),
    roll_opp_pts = rollmean(opp_pts, k=10, fill=NA, align='left'),
    roll_ptsdiff = rollmean(ptsdiff, k=10, fill=NA, align='left'))
```

Summary Data

```
team_box_summary <- team_box_scores_3 %>%
  group_by(season, team) %>%
  summarise(
    avg_fg_pct = round(mean(fg_pct, na.rm=TRUE), 3),
    avg_3fg_pct = round(mean(threefg_pct, na.rm=TRUE), 3),
    avg_pts = round(mean(pts, na.rm=TRUE), 1),
    avg_offrating = round(mean(offrating, na.rm=TRUE), 1),
    avg_defrating = round(mean(defrating, na.rm=TRUE), 1))
```

`summarise()` has grouped output by 'season'. You can override using the
`.groups` argument.

Data Export for Visualizations

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current')
saveRDS(team_box_scores_3, file = 'teamvisdata.rds')
saveRDS(team_box_summary, file = 'teamboxsummary.rds')
```